

Abstract

The aim of this project is to predict the heart diagnosis from ECG signals on the CPSC-2018 dataset. The dataset contains 6877 patients across 9 unique diagnosis codes. We will use machine learning and deep learning models trained on 12-lead ECG signals to classify each patient into one of the diagnostic categories. Model performance will be evaluated using metrics such as accuracy, F1 score, and ROC curve.

Description of the dataset

.mat files

Each patient has a .mat file which stores the raw ECG signal as a matrix. The number of samples and sampling frequency may vary between patients. This file contains the actual electrical activity of the heart across all 12 leads and will be used as input to the machine learning models. The file cannot be opened as a text file and requires a library such as wfdb to read it in Python.

.hea files

Table 1 – describes the basic technical parameters of the overall test, such as the total recording duration, the number of EKG leads used, and the date it was performed.

Table 2 - provides the instruction manual for the computer on how to read the raw numbers from each specific EKG cable and correctly scale them into an electrical voltage graph.

Table 3 - provides the patient's medical metadata, including their age, gender, and diagnosis.

Table 1.

ID	Identification number of this specific test
Number of Leads	Information about number of cables used in the test

Sampling Frequency	The number of times per second a continuous analog signal is measured.
Total Samples	Total number of data points
Data & Time	The exact date and time the patient was hooked up to the ECG machine.

Table 2.

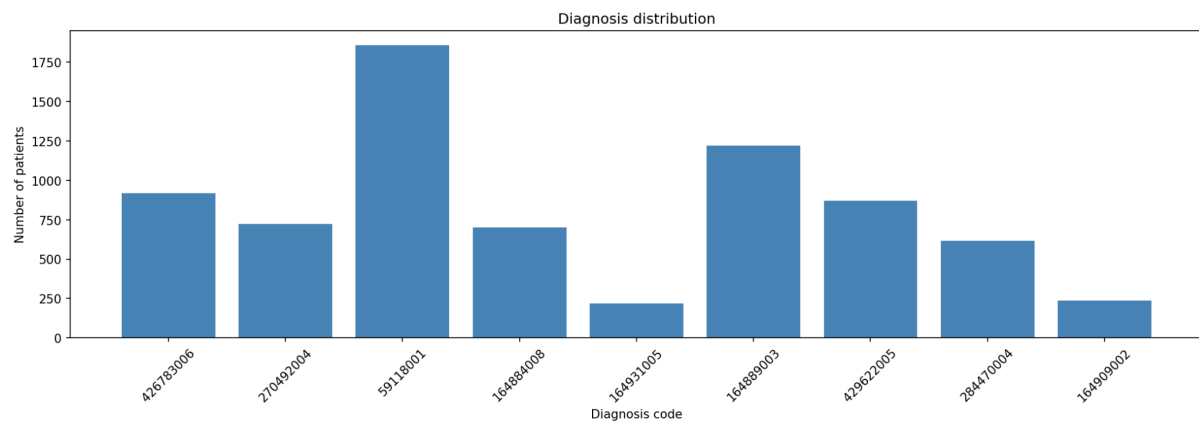
ID	Identification number of this specific test
Storage Format	how bits are laid out in memory.
Scale	The ratio defining how many digital units in the file equal 1 millivolt (1 mV) of real analog heart electricity.
ADC Resolution	The bit-depth of the analog-to-digital converter, indicating signal precision.
Baseline	The digital value that represents exactly zero electrical activity (the flat line).
Initial Value	The value of the very first data sample at the start of the recording.
Checksum	A mathematical value calculated over all samples to verify that the file is not corrupted.
Block Size	A formatting indicator defining whether the data stream is split into specific memory blocks.
Lead Label	The standard clinical name identifying which specific ECG electrode captured this signal.

Table. 3

Age	Age of the patient at the time of the recording.
Gender	The biological sex of the patient.
Diagnosis Code (Dx)	A standardized clinical terminology code (usually SNOMED CT) identifying the medical condition.
Medications (Rx)	Documented pharmaceutical treatments or drugs currently prescribed to the patient.
Medical History (Hx)	Past medical background, underlying conditions, or relevant patient history.

Symptoms (Sx)	Presenting clinical complaints or physiological symptoms reported by the patient.
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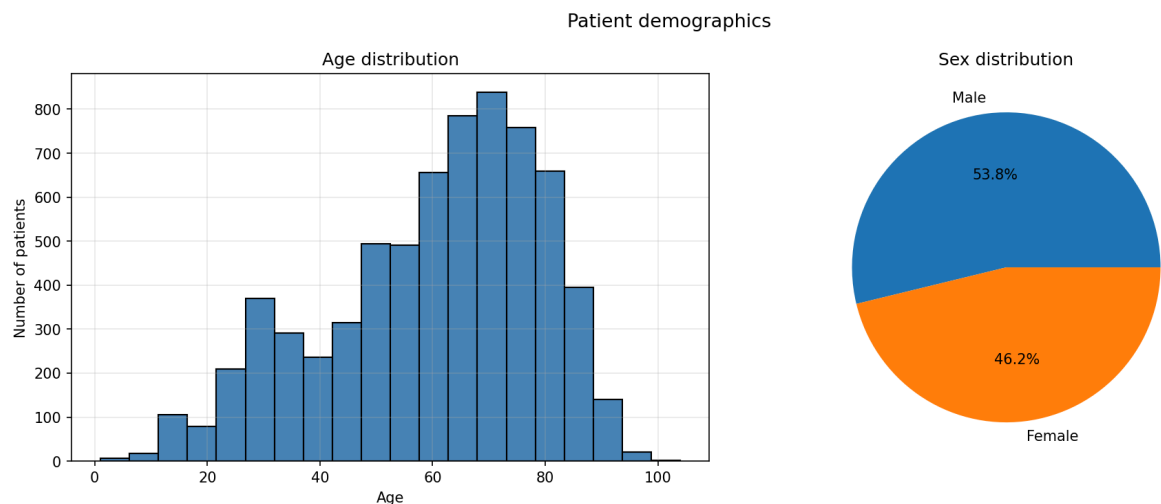
Diagnosis Distribution



Diagnosis (Dx)	patients (num)	
59118001	: 1857	The dataset contains 6877 patients across 9 unique diagnosis codes. The most frequent diagnosis is 59118001. The least common diagnosis is 164931005 with 220 cases. The dataset is notably imbalanced; the most common class has over 8x more samples than the rarest one. This will need to be addressed during model training, for example by using oversampling techniques or class weights in the loss function.
164889003	: 1221	
426783006	: 918	
429622005	: 869	
270492004	: 722	
164884008	: 700	
284470004	: 616	
164909002	: 236	
164931005	: 220	

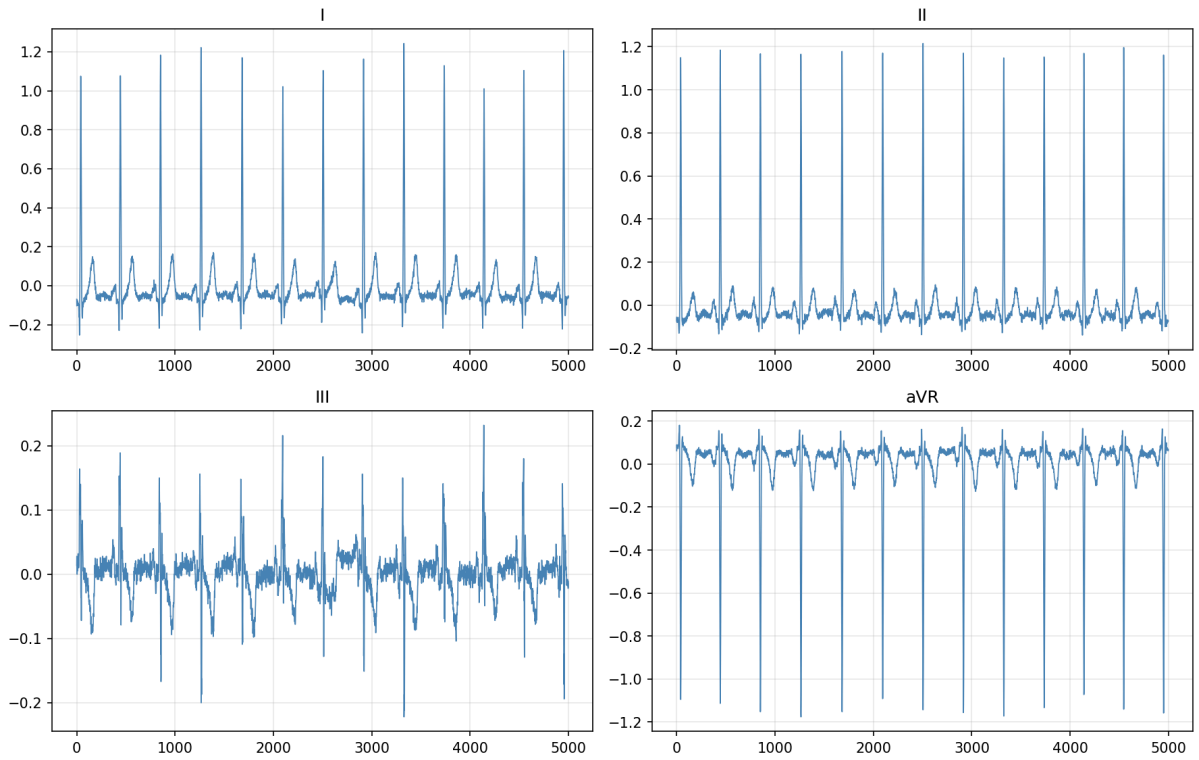
426783006 - (Normal Sinus Rhythm) is the only diagnosis code in this dataset that represents a healthy heart. All other 8 codes indicate various heart conditions such as atrial fibrillation, bundle branch blocks, or ventricular hypertrophy.

Patient Demographics

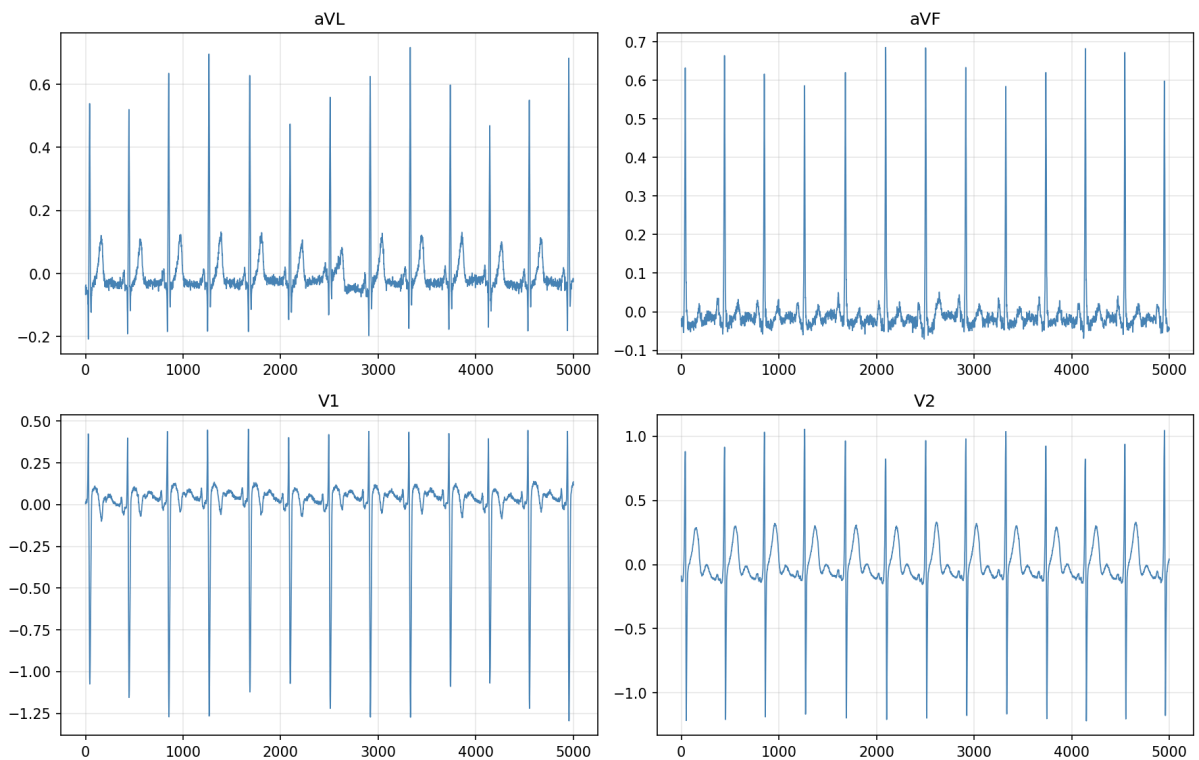


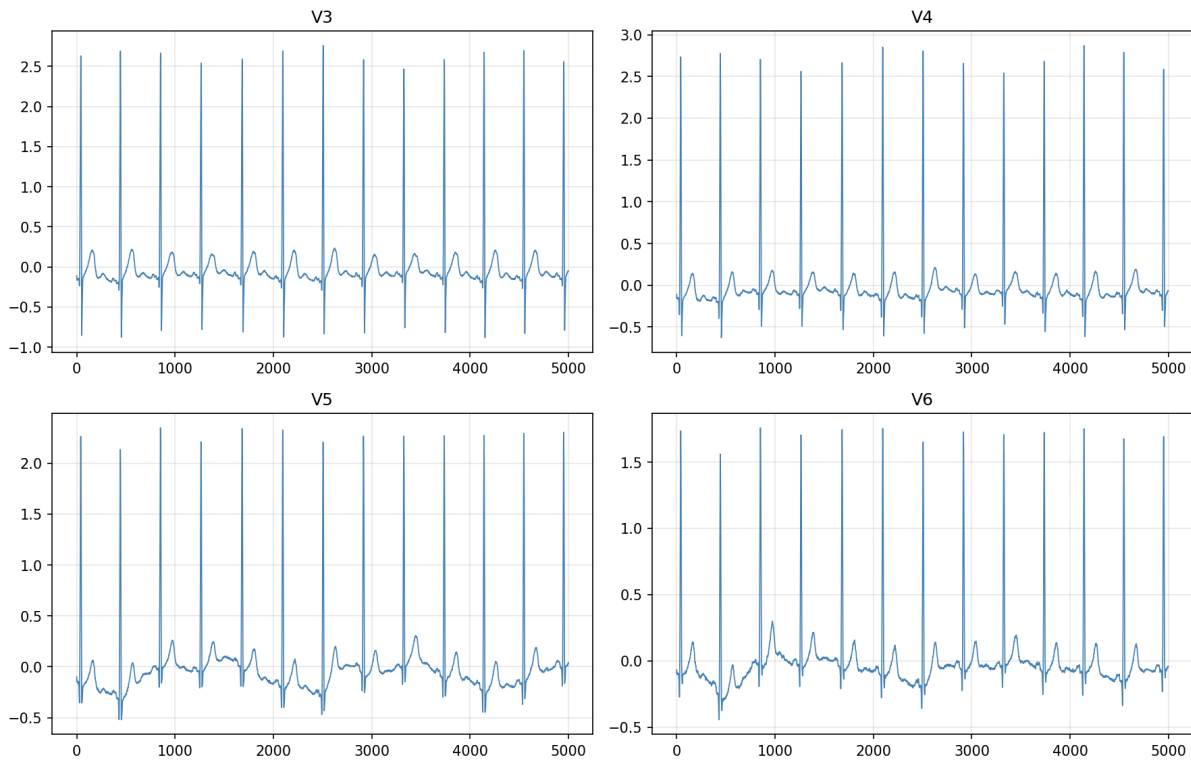
Age groups: 0-20: 189 21-40: 1024 41-60: 1730 61-80: 3004 81+: 921 Average age: 60.2	The dataset consists of 6877 patients with a fairly balanced sex distribution — 53.8% male and 46.2% female. The average age is 60,2 years, with the majority of patients falling in the 61-80 age group (3004 patients). The youngest group (0-20) is the least represented with only 189 patients, which reflects the fact that heart conditions are significantly more common in older populations.
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ECG: A2112 | Age: 55 | Sex: Male | Dx: 426783006 | Part 1/3



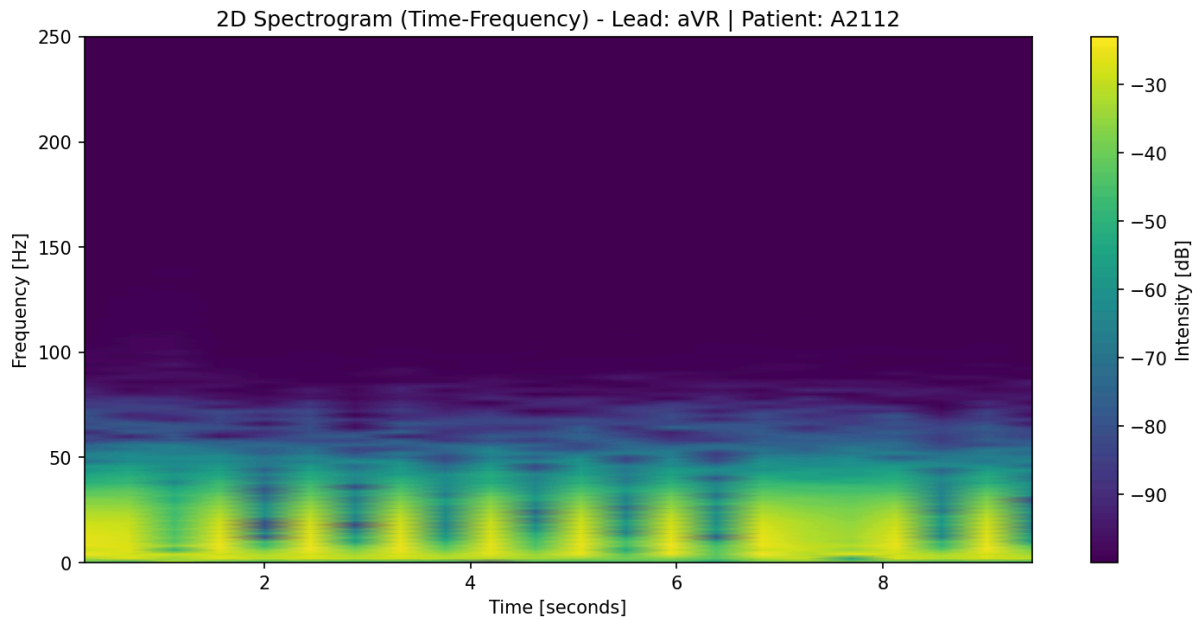
ECG: A2112 | Age: 55 | Sex: Male | Dx: 426783006 | Part 2/3





ECG Sample Visualization — Patient A2112

The figures show ECG signals of patient A2112, a 55-year-old male with diagnosis 426783006 (Normal Sinus Rhythm). The recording is split into 3 parts, each showing 4 out of 12 leads. Each lead records the electrical activity of the heart from a different angle. The signal spans 10 seconds with 5000 data points sampled at 500 Hz. The sharp peaks visible across all leads represent heartbeats, which appear regular and evenly spaced, which is expected for a patient with normal sinus rhythm.



2D Spectrogram (Time-Frequency) — Lead: aVR | Patient: A2112

The primary purpose of converting the 1D ECG signal into a 2D spectrogram is to extract both temporal (time) and spectral (frequency) features simultaneously. While 1D signals only show voltage changes over time, the 2D spectrogram exposes hidden frequency patterns—such as the rapid energy bursts of a heartbeat or the continuous low-frequency vibrations of an arrhythmia. This algebraic transformation turns a complex time-series problem into a computer vision task. It allows deep learning models (like 2D-CNNs) to classify heart conditions by recognizing distinct geometric shapes and color distributions instead of analyzing raw numerical rows.

Plan to solve the problem

Dataset Splitting Strategy

The dataset will be split into 80% training and 20% test set. The test set will be locked away and only used at the very end to evaluate the final model performance.

For model training and tuning we will use Stratified K-Fold Cross Validation.

Stratified version is chosen because the dataset is imbalanced — the most common class has over 8x more samples than the rarest one. Stratification ensures that each fold maintains the same ratio of diagnoses as the original dataset.

To further address the class imbalance we will apply SMOTE (Synthetic Minority Oversampling Technique) on the training set only. SMOTE creates new artificial samples for minority classes by interpolating between existing samples and their nearest neighbors, rather than simply duplicating them. This prevents the model from being biased towards the most common diagnosis.

Multi-Lead Data Integration and Proposed Classifiers

Since the CPSC-2018 dataset utilizes standard 12-lead ECG recordings, the models must process all 12 signals simultaneously to guarantee multi-class diagnostic accuracy. Before classification, each of the 12 denoised time-series leads is independently transformed into a 2D Mel-Spectrogram of size 128x128.

Algorithms

1. Baseline Classifier: Support Vector Machine (SVM)

Data Integration: All 12 spectrogram matrices are flattened individually into 1D arrays and concatenated horizontally into a single, comprehensive 1D feature vector per patient. This results in a final vector containing 196,608 unique features (12 leads multiplied by 128 multiplied by 128 pixels)

2. Custom 2D Convolutional Neural Network (2D-CNN)

Data Integration: The 12 individual spectrograms are stacked together along the channel dimension, creating a multi-channel tensor of shape (12, 128, 128) per patient. This acts exactly like a 12-channel image, where each channel represents a unique spatial view of the heart's electrical activity.

3. Deep Transfer Learning: ResNet-18

Data Integration: Similar to the Custom CNN, data is fed as a stacked (12, 128, 128) multi-channel tensor. The input layer of the standard network is modified to accept 12 channels instead of the standard 3 RGB channels.

Main Tools, Frameworks, and Libraries

WFDB (Waveform Database): Used to programmatically read and parse the clinical raw data matrices from the .mat files and extract relevant patient metadata from the corresponding .hea header files.

Scikit-learn: Used for splitting the dataset, implementing Stratified K-Fold cross-validation, and performing the SMOTE oversampling technique on the training set. It will also provide the Support Vector Machine (SVM) implementation and calculate core classification metrics

PyTorch / Keras: Used as the primary deep learning frameworks to construct, compile, and train the Custom 2D-CNN architecture as well as to load the pre-trained ResNet-18 model for transfer learning.

NumPy & Pandas: Employed for efficient high-dimensional array handling, matrix transformations, and restructuring structural patient metadata.

Matplotlib : Utilized to generate demographic charts, signal graphs, 2D spectrogram heatmaps, and final evaluation plots like the ROC curve.

SciPy: Applied during the preprocessing and time-frequency analysis stages. Specifically, `scipy.signal.butter` and `scipy.signal.filtfilt` will handle baseline wander removal, while `scipy.signal.spectrogram` will perform the short-time Fourier transformations.

Evaluation Methods and Performance Comparison

To strictly assess the performance of the three implemented classifiers and ensure a reliable multi-class diagnostic system, models will be validated using Stratified K-Fold Cross-Validation on the 80% training partition. The remaining 20% test set will be kept completely locked and utilized exclusively at the very end to compute the final, unbiased generalization performance.

Given that the CPSC-2018 dataset is highly imbalanced—with the dominant diagnosis appearing over eight times more frequently than the rarest one—relying solely on simple classification accuracy would be highly misleading. Therefore, a comprehensive set of classification metrics will be computed using a macro-averaged approach across all 9 diagnostic categories:

Accuracy: To measure the overall percentage of correctly predicted diagnostic categories across the entire evaluation set.

Specificity: To measure the model's ability to correctly identify true negative cases, ensuring that healthy patients or patients with distinct conditions are not misdiagnosed with the wrong defect.

[1] F. F. Liu *et al.*, "An open access database for evaluating the algorithms of ECG rhythm and morphology abnormal detection," *Journal of Medical Imaging and Health Informatics*, vol. 8, no. 7, pp. 1368–1373, 2018.

[2] A. L. Goldberger *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000, doi: 10.1161/01.CIR.101.23.e215.